

Undergraduate Medicine

Urology Study Pack

CharaNas Medicine Bot — Specialty Notes

Expanded notes with original schematic figures for revision. These are educational drawings inspired by standard undergraduate teaching themes, not copied textbook plates. Always follow your faculty curriculum and latest guidelines.

Section	Focus
1	Anatomy & stone disease
2	Infection spectrum
3	Hematuria & malignancy warnings
4	BPH, retention & scrotal emergencies
5	Checklist & books
6	Quick pearls from the bot notes
7	Recommended book sources

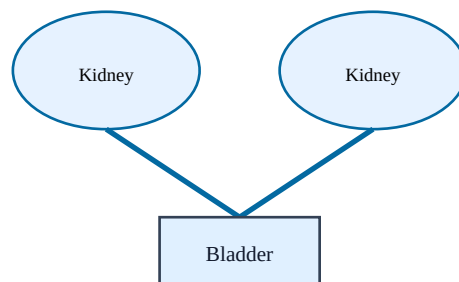
1. Anatomy & stone disease

Kidneys → ureters → bladder → urethra. Loin-to-groin pain suggests ureteric colic. Microscopic or gross hematuria often accompanies stones.

Stone types: calcium oxalate (most common), calcium phosphate, uric acid (often radiolucent on plain film), struvite (infection/urease organisms), cystine (genetic).

Non-contrast CT KUB is the usual definitive imaging in non-pregnant adults. Ultrasound is preferred first-line in pregnancy.

Figure: Urinary tract overview



Stone in ureter → loin-to-groin colic; obstruction + infection is an emergency

Schematic figure for learning — verify details in your recommended textbooks.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about urology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

2. Infection spectrum

Cystitis: dysuria, frequency, urgency, suprapubic discomfort — usually afebrile. Pyelonephritis: fever, flank pain, nausea — systemic illness.

Complicated UTI: male sex, obstruction, stones, catheters, immunosuppression, pregnancy, anatomic abnormality — needs careful culture and follow-up.

Obstruction + infection can rapidly progress to sepsis — urgent decompression may be required.

Revision prompts

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- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

3. Hematuria & malignancy warnings

Painless gross hematuria in older adults (especially smokers) raises concern for urothelial carcinoma of the bladder — needs urologic workup.

Renal cell carcinoma may present with hematuria, flank mass, or paraneoplastic features; many are incidental on imaging.

Always ask about anticoagulation, trauma, infection, stones, and constitutional symptoms.

Revision prompts

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- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

4. BPH, retention & scrotal emergencies

BPH causes storage and voiding LUTS in older men. Assess severity, infection, retention, creatinine, and medication effects.

Acute urinary retention: painful inability to void — catheterize and investigate cause. Post-obstructive diuresis can follow relief of chronic obstruction.

Testicular torsion: sudden severe pain in adolescent/young man — time-critical surgical emergency. Do not delay for imaging if high clinical suspicion.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about urology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

5. Checklist & books

Checklist: vitals, abdominal/flank/genital exam, UA $\pm$ culture, pregnancy test when relevant, imaging tailored to suspicion, analgesia for colic, sepsis pathway if shocked.

Books: Campbell-Walsh-Wein Urology; Smith's General Urology; Bailey & Love urology chapters.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about urology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

6. Quick pearls

High-yield reminders packaged with this specialty in the CharaNas bot:

1. Stone types: calcium oxalate most common; struvite with urease organisms; uric acid radiolucent.
2. Loin→groin pain + hematuria = ureteric colic until proven otherwise.
3. Painless hematuria: rule out malignancy (esp. smokers).
4. Pyelonephritis: fever + flank pain + UTI signs → antibiotics ± imaging.
5. BPH: storage/voiding symptoms in older men; assess PSA/exam judiciously.
 - Link symptoms → anatomy/physiology → investigation → first management step.
 - Write one OSCE-style explanation for a common presentation in this specialty.
 - After reading, do the bot Short MCQ and Case-based Question for active recall.

7. Recommended book sources

Standard undergraduate references commonly used internationally. Use the edition your college recommends. Figures in commercial textbooks are copyrighted; use library/licensed access for full plates and photographs.

#	Reference
1	Campbell-Walsh-Wein Urology
2	Smith's General Urology
3	Bailey & Love — Urology chapters

How to study with this PDF: read one section → sketch the figure from memory → answer bot MCQs/cases → review mistakes → revisit the matching section.

Disclaimer: educational aid only; not a substitute for clinical guidelines, faculty teaching, or licensed textbooks.